

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1738

Red Hill 345 kV
September 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1738. The Interconnection Customer proposes a 233.9 MW battery energy storage facility interconnecting at the Red Hill 345 kV substation in the Virginia area. The request was received on May 07, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 539 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,490,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHARLOTTESVILLE-MOHICAN 138.0 kV line	RED HILL-FLUVANNA P.S 345.0 kV L	141.8	168	30.9
UNISOURCE GOLDEN VLY-UNIONVILLE YDRO PROJECT 2499 NY 138.0 kV	LOUISA GENERATION FACILITY-FLUVANNA P.S 345.0 kV	131.3	266	5.2
BARRACKS ROAD-MOHICAN 138.0 kV line	RED HILL-FLUVANNA P.S 345.0 kV	126.6	200	30.9
TIMKEN MERCY-BARRACKS RD 138.0 kV	FLUVANNA P.S-RED HILL 345.0 kV Line	126.4	223	30.9
GORDONSVILLE ENERGY LP-LITTLEFORK 138.0 kV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345.0 kV	121.5	355	6.4
CHANCELLOR ROW-BEAR ISLAND 345.0 kV	LEAVITT-BIRCHWOOD 345.0 kV	107.1	2,328	7.8
CARSON ICE COGEN-ALDERWOOD 138.0 kV	CANDELARIA-MONEL 138.0 kV line	105.4	374	5.1
DUNN SIDING-BIRCHWOOD 345.0 kV	LEAVITT-BIRCHWOOD 345.0 kV line	103.5	1,519	7.0
FREDERICKSBURG-DUNN SIDIG 345.0 kV L	BIRCHWOOD-NAUVOO 345.0 kV L	100.5	1,914	6.2
WALWORTH-BLUFF ITY 138.0 kV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345.0 kV Line	100.0	413	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHARLOTTESILLE-MOHICAN 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,011,000	65% (\$1,298,000)
UNIONVILLE HYDRO PROJECT 2499 NY-UNISOURCE GOLDEN VALLEY 138.0 kV	Rebuild line with higher-capacity conductor	\$3,638,000	34% (\$1,223,000)
BARRACKS ROAD-MOHICAN 138.0 kV line	Rebuild line with higher-capacity conductor	\$8,597,000	65% (\$5,550,000)
BARRACKS RD-TIMKEN MERCY 138.0 kV	Rebuild line with higher-capacity conductor	\$6,175,000	65% (\$3,987,000)
GORDONSVILLE ENERGY LP-LITTLEFORK 138.0 kV	Rebuild line with higher-capacity conductor	\$2,848,000	27% (\$766,000)
CHANCELLOR ROW-BEAR ISLAND 345.0 kV	Reconductor line	\$10,569,000	1% (\$75,000)
CARSON ICE COGEN-ALDERWOOD 138.0 kV	Reconductor line	\$1,581,000	39% (\$620,000)
DUNN SIDING-BIRCHWOOD 345.0 kV line	Reconductor line	\$16,827,000	2% (\$328,000)
DUNN SIDING SUBSTATION-FREDERICKSBURG 345.0 kV	Reconductor line	\$9,900,000	1% (\$85,000)
WALWORTH-BLUFF ITY 138.0 kV	Reconductor line	\$3,855,000	66% (\$2,558,000)
	Total allocated to project		\$16,490,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.